

Chih-Wei Joshua Liu

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EDUCATION

Leland Stanford Junior University Biophysics PhD Program · Thesis advisor: Grant M. Rotskoff	Stanford, CA, USA 09/2024 – present
Massachusetts Institute of Technology (MIT) Bachelor of Science · Major: Physics · GPA: 4.9/5.0 · MCAT: 528/528	Cambridge, MA, USA 05/2022
National Experimental High School SAT: 1600/1600 · ACT: 36/36	Hsinchu, Taiwan, ROC 05/2018

SELECTED PROJECTS

C.-W.J. Liu, J. Klinger, G.M. Rotskoff, **Optimal parameterization of nonequilibrium generalized master equations from discrete-time experimental data**, [arXiv:2606.28289](#), in revision at *Phys. Rev. X* (2026). [\[code\]](#)

- Derived the first maximum-likelihood parameterization of master equations without Markov assumptions satisfying all physical constraints in general stochastic processes. Validated on three experimental systems.
- Set new state-of-the-art in benchmark extraction of HP35-Lys24Nle/Asn27His/Lys29Nle protein-folding kinetics from coarse-grained models trained on atomistic molecular-dynamics simulation.

J. Li*, **C.-W.J. Liu***, M. Szurek, and N. Fakhri, **Measuring irreversibility from learned representations of biological patterns**, *PRX Life* **2**, 033013 (2024), *equal contribution. [\[code\]](#)

- Extended thermodynamic-irreversibility estimation to entire experimental videos of high-dimensional biological processes using variational-autoencoder representations.

H. Lee*, E. Koskelo*, S. Gokhale*, J. Li*, C. Fei, **C.-W.J. Liu**, L. Lin, J. Dunkel, D.J. Skinner, and N. Fakhri, **Topological flowscape reveals state transitions in nonreciprocal living matter**, *Phys. Rev. X* **16**, 031061 (2026), *equal contribution.

- Trained classifier to distinguish particle classes with nonreciprocal interactions, enabling analyses of previously experimentally unobserved nonequilibrium states.
- Featured in *Physics* magazine and on the cover of *Phys. Rev. X*.

Change-point detection with convergence guarantee: [\[code\]](#)

- Developed `scikit-learn`-style Python interface for detecting change points in noisy time series, implementing the current state-of-the-art dynamic-programming-based algorithm.

POSITIONS

MIT Physics of Living Systems Group	Research Assistant (06/2022 – 08/2024)
MIT Department of Physics	Teaching Assistant, 8.01 Physics I (09/2019 – 12/2019)
	Teaching Assistant, 8.241 Introduction to Biological Physics (09/2023 – 05/2024)
Massachusetts General Hospital	Data Analysis Intern (06/2021 – 08/2021)
Analyzed patient statistics and authored two clinical publications.	

LEADERSHIP AND HONORS

National Science Foundation Graduate Research Fellowship (GRFP, 2024, Physics)

National Defense Science and Engineering Graduate Fellowship (2024, Physics, declined for GRFP)

Sigma Pi Sigma: Physics honor-society inductee; MIT Chapter President (2022) · **MIT News:** [profile \(2022\)](#)

USA Biology Olympiad: Top-50 High Honorist (2017); Semifinalist (2015–2018)

MIT Society of Physics Students: President (2021–2022)

Phi Kappa Theta – Massachusetts Eta: Pledgemaster (2021); sourced and fundraised donation of 12,000 surgical masks to Beth Israel Deaconess Medical Center during pandemic shortage as Service Chair (2019–2020).

SKILLS AND INTERESTS

Computing: lightning, jax, pydantic, numpy, scipy, torch, scikit-learn, statsmodels, uv, ty, ruff, modal
Natural language: Standard Chinese (native)

Research interests: nonequilibrium statistical mechanics, time series, optimal transport, LLM expressivity

Personal interests: phonology, historical linguistics, international-relations theory, Midwest emo, ichthyology